

Deflections of Beams

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ME 333: Mechanics of Materials

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I. INTRODUCTION

The objective of this experiment is to understand the behaviour of beams under bending loads and test the predictions of Euler-Bernoulli beam theory.

A. Assumptions of the Euler-Bernoulli Beam Theory

The assumptions on the deformation of a beam that the Euler-Bernoulli theory makes are as follows:

- 1) Plane sections remain planar and perpendicular to the neutral axis, which ensures that the shear deformations are negligible.
- 2) The deflections are small, which leads to the slope of the neutral axis also being small, and this helps us neglect the term dv/dx in the denominator, simplifying the relation.
- 3) The material follows Hooke's Law, meaning that stress is proportional to strain.
- 4) The material is a linear elastic material, and after the load is removed, it returns to its original position.

B. Deflected Shape of the Cantilever Beam

For a beam with Young's modulus E and second moment of area I , loaded with uniformly distributed load q along its length, the deflection y from a straight unloaded configuration is governed by the equation:

$$\frac{d^4 y}{dx^4} = \frac{q}{EI}$$

Rearranging the terms we get the following relation.

$$\frac{d^2}{dx^2} \left(EI \frac{d^2 y}{dx^2} \right) = q$$

$$\frac{d^2 M(x)}{dx^2} = q$$

Substituting the relation for q we get the following equation

$$EI \frac{d^2 y}{dx^2} = M(x)$$

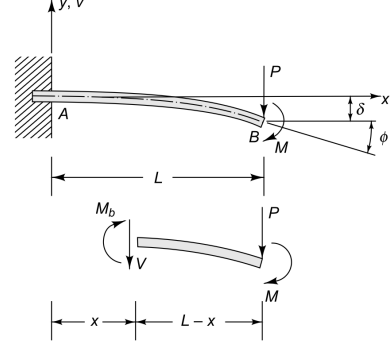


Fig. 1: Free body diagram of an element (Crandall 3rd ed.)

To calculate the deflection we calculate the moment at a distance x from the fixed end. The negative sign indicates the load is acting in the downward direction.

$$M(x) = -P(L - x)$$

Substituting $M(x)$ into the beam equation:

$$EI \frac{d^2 y}{dx^2} = -P(L - x)$$

Integrating twice with respect to x : with the boundary conditions being the slope (dy/dx) at $x = 0$ and the deflection at $x = 0$ being 0 at the fixed end, we get the following expression for deflection $y(x)$

$$y(x) = -\frac{P}{6EI} x^2 (3L - x)$$

C. Bending Moment and Shear Force Expressions

To calculate the bending moment in the cantilever beam we can use our original bending moment function derived.

$$M(x) = -P(L - x)$$

To calculate the shear force in the cantilever beam we can use our bending moment function derived and use the following relation between moment and shear force given below.

$$\frac{dM}{dx} + V(x) = 0$$

Substituting the moment function in the relation we get the following shear force function.

$$V(x) = -P$$

The shear force in the cantilever beam is constant throughout the beam and equal to the magnitude of the force applied.

II. PROCEDURE

- 1) Place the rectangular beam of the given material on the beam deflection apparatus and tighten the supports such that the beam is firm and does not sag noticeably under its own weight.
- 2) Place the digital indicator (for deflection measurement) at the point where the load will be placed ($0.25L$, $0.50L$, $0.75L$, L). Note the initial reading of the indicator.
- 3) Now, place the specified load on the load hanger and then place the hanger at the specified position on the beam.
- 4) Record the deflection indicated by the digital indicator at that position.
- 5) For the same load, move the load to other positions and slide the digital indicator to the corresponding points.
- 6) Repeat the above steps for five different loads.
- 7) Repeat the entire procedure for beams made of different materials.

III. OBSERVATIONS

In this section, we present the experimentally observed and theoretically calculated deflection of a beam (at the point of application of load) made of three materials: Aluminium, Stainless Steel, and Brass, for five different point loads applied at four locations: $0.25L$, $0.50L$, $0.75L$, and L , where L is the total length of the cantilever beam.

Using Euler-Bernoulli Beam Theory, the deflection of a cantilever beam (beam with one end fixed, the other free) at the location of application of a point load is given by:

$$y_l = \frac{Pl^3}{3EI} \quad (1)$$

Where P is the load (force), l is the distance from the fixed end to the point of application of the load, E is the Young's modulus of the material, and I is the second moment of cross-sectional area about the axis, about which the beam bends/deflects.

In this experiment:

- 1) P (load in force) is given by: $P = mg$, where m is the hanging mass at the application point and g is the acceleration due to gravity.
- 2) l (distance from fixed end to application point) are 75 mm, 150 mm, 225 mm and 300 mm (total beam length L is set to 300 mm).
- 3) E (Young's modulus) of the three materials were given as (by the experimental setup): $E_{Al} = 69$ GPa, $E_{ss} = 207$ GPa, $E_{br} = 105$ GPa
- 4) I (second moment of cross-sectional area) is given by: $I = bh^3/12$ where b is the width and h is the height of the beam. In the experiment, $b = 19$ mm, $h = 3$ mm and therefore $I = 42.75 \text{ mm}^4$.

The experimentally observed data for the three materials are presented in both tabular and graphical formats as follows.

TABLE I: Aluminium: Experimental Deflection (in mm)

Distance (in mm) Load (in g)	75	150	225	300
40	0.02	0.10	0.67	1.21
80	0.05	0.21	1.08	2.21
120	0.05	0.44	1.44	3.42
160	0.11	0.60	1.91	4.39
200	0.12	0.78	2.47	5.27

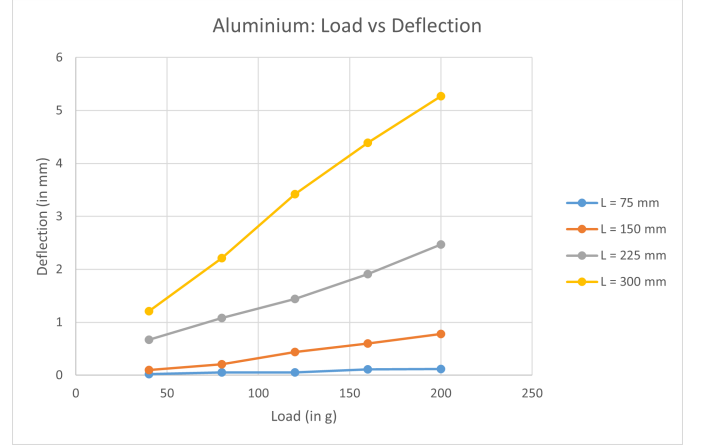


Fig. 2: Load vs Deflection plot for different locations

TABLE II: Stainless Steel: Experimental Deflection (in mm)

Distance (in mm) Load (in g)	75	150	225	300
50	0.01	0.07	0.36	0.57
100	0.02	0.10	0.54	0.88
150	0.03	0.14	0.74	1.60
200	0.04	0.23	0.89	1.78
230	0.05	0.26	1.10	2.45

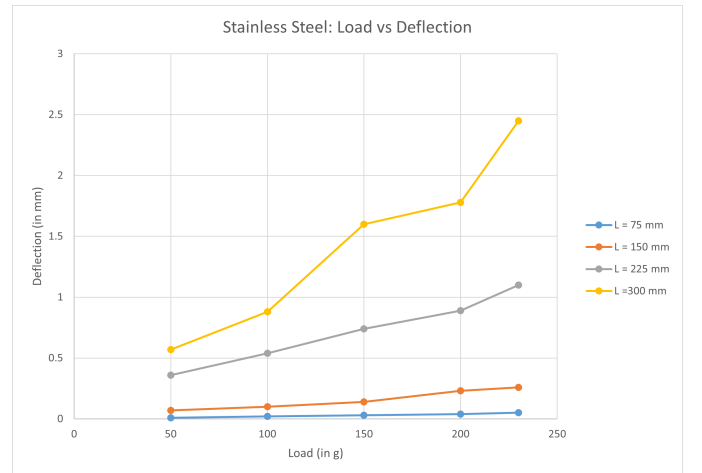


Fig. 3: Load vs Deflection plot for different locations

TABLE III: Brass: Experimental Deflection (in mm)

Distance (in mm) Load (in g)	75	150	225	300
100	0.05	0.37	1.02	2.57
140	0.04	0.52	1.43	3.36
160	0.08	0.60	2.21	3.81
200	0.09	0.73	1.99	4.74
230	0.11	0.91	2.30	5.46

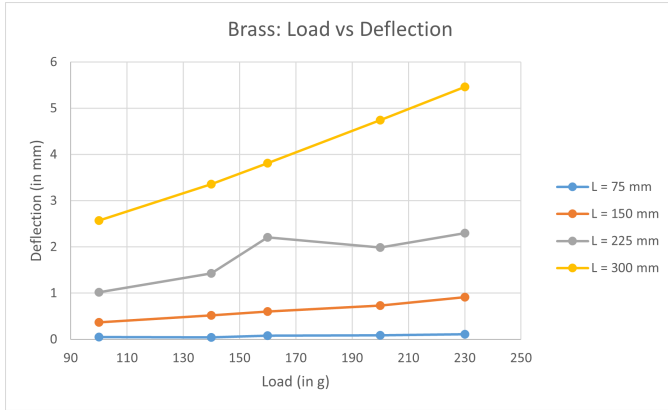


Fig. 4: Load vs Deflection plot for different locations

Using the parameters presented earlier and the equation (1), the theoretical deflection is calculated and presented in the Tables IV, V and VI as follows.

TABLE IV: Aluminium: Theoretical Deflection (in mm)

Distance (in mm) Load (in g)	75	150	225	300
40	0.0187	0.1496	0.5051	1.197
80	0.0371	0.2993	1.0102	2.3945
120	0.0561	0.449	1.515	3.5918
160	0.0748	0.5986	2.0204	4.789
200	0.0935	0.7483	2.525	5.9863

TABLE V: Stainless Steel: Theoretical Deflection (in mm)

Distance (in mm) Load (in g)	75	150	225	300
50	0.00779	0.06236	0.21045	0.49886
100	0.0156	0.12471	0.42091	0.99771
150	0.0234	0.18707	0.63136	1.49657
200	0.0311	0.24943	0.84182	1.99542
230	0.0358	0.28684	0.96809	2.29474

TABLE VI: Brass: Theoretical Deflection (in mm)

Distance (in mm) Load (in g)	75	150	225	300
100	0.0307	0.2459	0.8298	1.967
140	0.0430	0.3442	1.1617	2.7537
160	0.0492	0.3934	1.3277	3.1471
200	0.0615	0.4917	1.6596	3.9338
230	0.0707	0.5655	1.9085	4.5239

The error (in %) in experimental value with respect to the theoretical value (Error = (Experiment Value – Theoretical Value)/Theoretical Value $\times$ 100) is also tabulated

in the Tables VII, VIII and IX, as follows (The link of the (excel) file containing experimental data, theoretically computed values, errors and plots are attached the end of the report).

TABLE VII: Aluminium: Error (in %)

Distance (in mm) Load (in g)	75	150	225	300
40	6.91	33.18	32.65	1.06
80	33.64	29.84	6.91	7.71
120	10.91	1.99	4.97	4.78
160	47.00	0.23	5.46	8.33
200	28.29	4.24	2.20	11.97

TABLE VIII: Stainless Steel: Error (in %)

Distance (in mm) Load (in g)	75	150	225	300
40	28.29	12.26	71.06	14.26
80	28.29	19.82	28.29	11.80
120	28.29	25.16	17.21	6.91
160	28.29	7.79	5.72	10.80
200	39.45	9.36	13.63	6.77

TABLE IX: Brass: Error (in %)

Distance (in mm) Load (in g)	75	150	225	300
40	62.69	50.49	22.92	30.66
80	7.03	51.07	23.09	22.02
120	62.69	52.52	66.46	21.07
160	46.42	48.46	19.91	20.49
200	55.62	60.92	20.51	20.69

Observation & Interpretation: First, we observe that the deflection at the load application point (y_l) increases linearly with the point load (P), but the slope of the lines increase rapidly (non-linearly) with distance of location of load application point (l), as seen in figures 2, 3 and 4. This is evident from the equation (1), where $y_l \propto P$ and the slope ($= l^3/3EI$) $\propto l^3$. Consequently, we also note that for a given load P , the deflection y_l increases non-linearly with distance l , as per equation (1) where $y_l \propto l^3$.

Therefore, the experimental observation aligns well with the theoretical formulation. For aluminium, the errors are as small as 0.23 % and go at most 47 %. However, most error percentages are around 1 - 10 %. Similarly, for stainless steel and brass, though errors are higher than in aluminium.

IV. SOURCES OF ERROR

- 1) The theoretical analysis was done using the standard Modulus of elasticity, assuming the experimental beam had the same value.
- 2) The digital indicator may not be properly calibrated, leading to a systematic offset in deflection readings.
- 3) Slight slack or looseness in the setup might have contributed to additional deflection, increasing measurement error.
- 4) Parallax error during the positioning of the load and digital indicator may also have affected the readings.

V. CONCLUSION

The experiment confirms that beam deflection increases linearly with load and nonlinearly with the distance of load application, consistent with Euler–Bernoulli beam theory. The experiment successfully helped to understand the theoretical relationship between load, position, and deflection for elastic beams.

DATA LINK

[https://github.com/MukeshK17/
ME333-Deflections-in-elastic-beams.git](https://github.com/MukeshK17/ME333-Deflections-in-elastic-beams.git)

VI. REFERENCES

- 1) “Plane Beam Approximations: Euler Bernoulli Beam ,” Engineering at Alberta, <https://engcourses-uofa.ca/books/introduction-to-solid-mechanics/plane-beam-approximations/euler-bernoulli-beam/> (accessed Nov. 1, 2025).
- 2) S. H. Crandall, N. C. Dahl, and T. J. Lardner, *An Introduction to the Mechanics of Solids*, 3rd ed.